VINUNIVERSITY

Test duration: 1 hour and 30 minutes

Name: **Student ID:**

Midterm Practice Test 1 – Fall 2026

COMP3040/5030 Computer Vision Unofficial practice test, prepared by Lê Tăng Phú Quý

Instructions:

- Academic honesty is required in all work you submit to be graded. You should **NOT** copy or share your solutions with other students to avoid plagiarism issues.
- For short answer questions, please make your answers as **concise** as possible.
- This is a closed-book exam and NO Internet access is allowed. You are allowed 1 page of **hand-written** note (both sides).
- Drafting paper and pen/pencil are allowed. We will provide the drafting paper to you.
- Late submission without an approved extension will not be graded at all.
- You must **RETURN** this hard copy of the midterm exam at the end of the exam period.
- During the exam, you are **NOT** allowed to leave the room or/and use the restroom.
- You are required to **NOT** use your mobile phone in the exam room. Please put it on the table or in your bag.
- Good luck!

About the exam: There are **05** problems with **100 points** total, and some have multiple parts. Please check through the pages before you start.

Problem	Content	Point (Undergraduates)	Type
1	Fundamentals of Computer Vision	20	short answer
2	Image Filtering	25	long answer
3	Feature Detection and Matching	25	long answer
4	Fitting and Image Transformations	25	long answer
5	System Design	05	long answer
Total		**100**	

Problem 1. Fundamentals of Computer Vision

Give a short answer for each of the following question:

1.1. What is the difference between image **filtering** and image **warping**? Give one example of each. (5 pts)

1.2. Write the **negative** transformation for an 8-bit grayscale image and for an image normalized to $[0, 1]$. Why is it a *point operator*? (5 pts)

1.3. Name the four main steps of SIFT. (5 pts)

1.4. Why do we need **homogeneous coordinates** to represent affine transformations? (5 pts)

Problem 2. Image Filtering

2.1. A 5×5 image has $I[i, j] = (i + 1)(j + 1)$ (row i, column j, from 0). It is filtered with a 3×3 **box** (mean) filter. (a) Output size with padding 0, stride 1? padding 1, stride 1? padding 1, stride 2? (b) The value at the centre $(2, 2)$. (c) The value at the corner $(0, 0)$ with zero padding and with replicate padding; comment. (12 pts)

2.2. A 3×3 window holds $\{12, 15, 255, 14, 13, 0, 16, 11, 14\}$. Compute the output of a **mean** filter and of a **median** filter. What kind of noise is this, and which filter should be used? (8 pts)

2.3. Is the median filter linear? Prove it or give a counterexample. (5 pts)

Problem 3. Feature Detection and Matching

3.1. For the patch $\begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 50 \\ 10 & 50 & 50 \end{bmatrix}$ and the (correlation-form) Sobel kernels $S_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$, $S_y = S_x^\top$: (a) compute I_x, I_y at the centre; (b) the gradient magnitude and direction (y axis pointing down); (c) in Canny's NMS, which two neighbours is the centre compared with? (13 pts)

3.2. Template matching with SSD, NSSD and ZNCC. (a) Write the formulas. (b) If the source image is re-taken in darker light (every pixel $I \to 0.6I + 10$), which measure is still reliable, and why? (c) Why does template matching fail when the object is rotated by 30°? (12 pts)

Problem 4. Fitting and Image Transformations

4.1. Fit $y = mx + b$ to $(1, 1), (2, 3), (3, 2), (4, 4)$ by least squares. Write the normal equations, find m, b and the sum of squared errors. (13 pts)

4.2. (a) Rotate $(2, 1)$ by $90°$ counter-clockwise about the origin (math axes, y up), then translate by $(3, -1)$. Write the 3×3 homogeneous matrix and the result. (b) Redo in the opposite order. What do you conclude? (12 pts)

Problem 5. System Design

5. You build a **prescription digitizer**: users photograph a paper prescription with a phone (tilted, skewed, cluttered desk). Design an automatic pipeline that turns the photo into a canonical, front-facing, fixed-size image of the page, ready for OCR. (5 pts)